COMP 441 — Software Analysis and Testing

Week 6 — AI Baseline, Coverage and Static Analysis

Group A · Track D · TOML Kit

Period	Week 6
Evidence folder	week6
Repository	github.com/the-dezeray/tomlkit
Commit(s)	8c959b5
Generated	2026-10-27
Contents	12 sections

This pack collects the week's human-readable report and key evidence. Raw AI-only logs are listed in the inventory and intentionally not printed.

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Source: week6\README.md

Week 6 evidence (Group A, Track D, tomlkit @ 8c959b5)

Run everything from the repo root.

Reproduce

```
python -m venv .venv && . .venv/bin/activate
pip install -e . pytest pytest-cov radon pylint ruff crosshair-tool
git submodule update --init

# AI-only baseline (expect 385 passed, 3 xfailed; 81.7% stmt / 66.3% branch)
pytest -q --cov=tomlkit --cov-branch --cov-report=html:week6/htmlcov_ai tests_ai
# Existing suite, separate run (expect 1058 passed; 89.5% / 83.1%)
pytest -q --cov=tomlkit --cov-branch tests

python week6/scripts/assert_audit.py tests_ai week6/assertion_audit.csv
python week6/scripts/gap_by_function.py week6/cov_ai.json . 25

radon cc tomlkit -s -n C --total-average
radon mi tomlkit -s
pylint tomlkit --disable=C0114,C0115,C0116
ruff check tomlkit --select ALL --statistics

cd week6/symbolic && crosshair check ch_date.py --per_condition_timeout=60
```

Files

- `tests_ai/` — AI-generated suite + `PROMPT_LOG.md` (prompts, iterations)
- `rtm/rtm_week6.csv` — traceability matrix (16 requirements); `build_rtm.py` validates test refs
- `assertion_audit.csv` — per-test assertion verdict
- `htmlcov_ai/`, `cov_*.json` — coverage
- `radon_*`, `pylint.txt` — static analysis
- `symbolic/` — CrossHair contract + counterexample for defect D1

Requirements traceability matrix (R01–R16)

Source: week6\rtm\rtm_week6.csv

Req ID	Requirement	Source	Risk priority	AI-baseline tests (tests_ai/)	Status after AI baseline	Planned tests (level, week)
R01	Parse any valid TOML 1.1 document into a TOMLDocument whose values match the spec types	README ("1.1.0-compliant"); CHANGELOG 0.15.0 #456	High	test_ai_parser.py::TestKeys; test_ai_parser.py::TestStrings; test_ai_parser.py::TestIntegers; test_ai_parser.py::TestFloats; test_ai_parser.py::TestDates; test_ai_parser.py::TestTables; test_ai_parser.py::TestArrayOfTables	Covered (unit)	Integration: run toml-test valid corpus through parse→unwrap (Wk 9); Property: Hypothesis value round-trip (Wk 7)
R02	parse() then dumps() on an unmodified document returns the identical string (comments, whitespace, order preserved)	README ("preserves all comments, indentations, whitespace and internal element ordering")	High	test_ai_api.py::TestRoundTrip; test_ai_parser.py::TestArrays::test_multiline_trailing_comma_and_comments; test_ai_parser.py::TestWhitespaceAndNewlines::test_crlf_newlines	Covered (unit, 8 fixed examples)	Property: Hypothesis parse→dumps identity on generated docs (Wk 7); BDD scenario 'Unchanged file is byte-identical' (Wk 8)
R03	Editing a parsed document changes only the edited value; untouched comments and layout are preserved	README ("accessible and editable via an intuitive API")	High	test_ai_container.py::TestEditingPreservesStyle; test_ai_items.py::TestIntegerFloat::test_integer_keeps_trivia_comment; test_ai_toml_file.py::TestTOMLFile::test_edit_and_write	Partial: Container.append/_insert_after/_replace_at largely unexecuted (see coverage gaps)	Manual unit tests from Wk 7 surviving mutants; BDD 'Bump a version keeps comments' (Wk 8)
R04	Malformed input raises a ParseError subclass that reports the line and column of the problem	Code: exceptions.py, parser.py (parse_error)	High	test_ai_parser.py::TestErrorPositions; test_ai_parser.py::TestStrings::test_control_char_rejected; test_ai_parser.py::TestKeys::test_missing_key	Partial: ~25 negative tests assert only ParseError; D3 inline duplicate key escapes ParseError	Manual: assert exact subclass + line/col per error type (decision table, Wk 7); integration with toml-test invalid corpus (Wk 9)
R05	dumps() serialises Mapping/Table/Container input, honours sort_keys, and raises TypeError for anything else	Docstring: api.dumps; CHANGELOG #482	Medium	test_ai_api.py::TestDumps; test_ai_api.py::TestEncoders::test_register_and_unregister	Covered (unit); sort_keys ignored on wrapper path not tested	Manual: wrapper + sort_keys case (Wk 7)

Req ID	Requirement	Source	Risk priority	AI-baseline tests (tests_ai/)	Status after AI baseline	Planned tests (level, week)
R06	load()/dump() work with text and binary file-like objects	Docstring: api.load, api.dump	Medium	test_ai_api.py::TestFileLikeIO	Covered (unit)	Integration: dump→file→load round trip with real file (Wk 9)
R07	TOMLFile.read/write round-trips a file and preserves LF, CRLF and mixed line endings	Code: toml_file.py	Medium	test_ai_toml_file.py::TestTOMLFile	Covered (unit, 98% stmt)	Integration test with real filesystem + edit (Wk 9)
R08	date()/time()/datetime() factories return the right item type and reject strings of another kind	Docstring: api.date ("only accepts date strings")	Medium	test_ai_api.py::TestDateTimeFactories	FAILS: D1 date() accepts datetime strings (xfail)	Fix via TDD cycle #1 (Wk 8); Hypothesis over generated datetime strings (Wk 7)
R09	item() converts supported Python types to TOML items, raises ConvertError otherwise, and consults registered custom encoders	Docstring: items.item, api.register_encoder	Medium	test_ai_items.py::TestItemConversion; test_ai_api.py::TestEncoders	Covered (unit); many checks are type-only	Property: Hypothesis item(x).unwrap() == x for generated JSON-like data (Wk 7)
R10	string() applies TOML escaping for basic strings, supports literal/multiline forms, and rejects sequences invalid for the chosen form	Docstring: api.string; TOML spec §String; CHANGELOG (leading-newline fix)	Medium	test_ai_api.py::TestStringFactory; test_ai_items.py::TestStringFromRaw; test_ai_utils.py::TestEscapeString	Covered (unit)	Property: string(s) re-parses to s for arbitrary text (Wk 7)
R11	Arithmetic on Integer/Float/Date/DateTime items returns TOML items of the same kind and keeps trivia	Code: items.py (Integer.__add__ etc.)	Low	test_ai_items.py::TestIntegerFloat; test_ai_items.py::TestDates	Covered (unit)	—
R12	Array editing (append/insert/delete/set/add_line) yields valid, re-parseable output; slice assignment is rejected	Docstring: Array.add_line; CHANGELOG #499, #527; commit #600	High	test_ai_items.py::TestArray	Partial: Array.__delitem__ and insert edge paths unexecuted	Manual BVA tests on index 0 / -1 / len; state-transition tests for comma placement (Wk 7)
R13	Out-of-order and dotted-key tables can be read, edited and deleted; re-serialisation follows the documented limitation	README §Limitations; CHANGELOG #261, #536, #545	High	test_ai_container.py::TestDottedAndOutOfOrder; test_ai_parser.py::TestTables::test_out_of_order_tables_round_trip	Partial: OutOfOrderTableProxy.__setitem__ / _merge_aot_fragment unexecuted	Integration: proxy + container + renderer scenarios (Wk 9)
R14	Comments created through the API always produce valid TOML; line breaks are rejected or split	Docstring: api.comment; CHANGELOG #449, commit #544	Medium	test_ai_api.py::TestTriviaFactories::test_multiline_comment; test_ai_items.py::TestItemDecorations::test_comment_rejects_newline; test_ai_items.py::TestArray::test_add_line_comment_with_newline_rejected	FAILS for \r: D6 comment('a\rb') writes unparseable TOML (no AI test caught it)	Manual: \r and \r\n cases (Wk 7); TDD fix (Wk 8)
R15	Keys built with key() always serialise to a valid, re-parseable key	Docstring: api.key; reverse-engineered from code	Low	test_ai_api.py::TestKeyFactory	FAILS: D5 key([]) serialises as ' = 1' (no AI test caught it)	Manual BVA: empty / one / many segments (Wk 7)
R16	Parsing time grows roughly linearly with document size (no quadratic blow-up)	CHANGELOG 0.15.1 #479, #503, #517, #525 (performance fixes); Risk R5	Medium	—	UNTESTED by AI suite	Light-touch performance test with Locust/pytest-benchmark on large lock-file style input (Wk 9)

Radon cyclomatic complexity (rank C and above)

Source: week6\radon_cc.txt

```
tomlkit/container.py
    M 225:4 Container.append - F (74)
    M 871:4 Container._replace_at - F (41)
    M 661:4 Container._render_table - E (31)
    M 1173:4 OutOfOrderTableProxy.__setitem__ - C (16)
    M 420:4 Container._validate_table_candidate - C (15)
    M 563:4 Container._insert_at - C (15)
    F 1272:0 _equal_with_nan - C (13)
    M 516:4 Container._insert_after - C (13)
    M 153:4 Container._get_last_index_before_table - C (12)
    M 633:4 Container.as_string - C (11)
tomlkit/parser.py
    M 867:4 Parser._parse_string - E (40)
    M 1006:4 Parser._parse_table - E (37)
    M 480:4 Parser._parse_value - D (25)
    M 292:4 Parser._parse_comment_trail - D (23)
    M 744:4 Parser._parse_number - D (22)
    M 633:4 Parser._parse_array - C (13)
    M 175:4 Parser.parse - C (12)
    M 811:4 Parser._parse_escaped_char - C (12)
tomlkit/_utils.py
    F 48:0 parse_rfc3339 - C (11)
tomlkit/items.py
    M 1655:4 Array.__delitem__ - E (36)
```

```
    F 114:0 item - E (31)
    M 2055:4 InlineTable.as_string - D (30)
    M 1595:4 Array.insert - D (26)
    M 1493:4 Array.add_line - D (24)
    M 2316:4 AoT.insert - C (15)
    M 1395:4 Array._group_values - C (11)

524 blocks (classes, functions, methods) analyzed.
Average complexity: A (3.0381679389312977)
```

Radon maintainability index

Source: week6\radon_mi.txt

```
tomlkit/container.py - C (0.00)
tomlkit/toml_file.py - A (74.02)
tomlkit/_compat.py - A (66.26)
tomlkit/__init__.py - A (100.00)
tomlkit/parser.py - C (0.00)
tomlkit/_types.py - A (80.64)
tomlkit/source.py - A (57.62)
tomlkit/api.py - A (64.23)
tomlkit/exceptions.py - A (68.60)
tomlkit/toml_document.py - A (100.00)
tomlkit/_utils.py - A (52.71)
tomlkit/items.py - C (0.00)
```

Pylint output

Source: week6\pylint.txt

```
************* Module tomlkit.container
tomlkit/container.py:1: [C0302(too-many-lines), ] Too many lines in module (1287/1000)
tomlkit/container.py:139: [W0212(protected-access), Container._handle_dotted_key] Access to a protected member _dotted o
f a client class
tomlkit/container.py:142: [W0212(protected-access), Container._handle_dotted_key] Access to a protected member _dotted o
f a client class
tomlkit/container.py:135: [R1711(useless-return), Container._handle_dotted_key] Useless return at end of function or met
hod
tomlkit/container.py:225: [R0914(too-many-locals), Container.append] Too many local variables (19/15)
tomlkit/container.py:274: [R1705(no-else-return), Container.append] Unnecessary "elif" after "return", replace only that
 "elif" with "if"
tomlkit/container.py:261: [R1702(too-many-nested-blocks), Container.append] Too many nested blocks (6/5)
tomlkit/container.py:304: [R1705(no-else-return), Container.append] Unnecessary "elif" after "return", replace only that
 "elif" with "if"
tomlkit/container.py:261: [R1702(too-many-nested-blocks), Container.append] Too many nested blocks (6/5)
tomlkit/container.py:389: [R1705(no-else-return), Container.append] Unnecessary "else" after "return", remove the "else"
 and de-indent the code inside it
tomlkit/container.py:400: [W0212(protected-access), Container.append] Access to a protected member _previous_item of a c
lient class
tomlkit/container.py:225: [R0911(too-many-return-statements), Container.append] Too many return statements (9/6)
tomlkit/container.py:225: [R0912(too-many-branches), Container.append] Too many branches (46/12)
tomlkit/container.py:225: [R0915(too-many-statements), Container.append] Too many statements (92/50)
tomlkit/container.py:425: [W0212(protected-access), Container._validate_table_candidate] Access to a protected member _m
ap of a client class
tomlkit/container.py:445: [W0212(protected-access), Container._validate_table_candidate] Access to a protected member _m
ap of a client class
tomlkit/container.py:451: [W0212(protected-access), Container._validate_table_candidate] Access to a protected member _m
ap of a client class
tomlkit/container.py:661: [R0912(too-many-branches), Container._render_table] Too many branches (16/12)
tomlkit/container.py:871: [R0914(too-many-locals), Container._replace_at] Too many local variables (16/15)
tomlkit/container.py:871: [R0912(too-many-branches), Container._replace_at] Too many branches (17/12)
tomlkit/container.py:871: [R0915(too-many-statements), Container._replace_at] Too many statements (53/50)
tomlkit/container.py:994: [W0613(unused-argument), Container._getstate] Unused argument 'protocol'
tomlkit/container.py:1077: [W0212(protected-access), OutOfOrderTableProxy.validate] Access to a protected member _body o
f a client class
tomlkit/container.py:1083: [W0212(protected-access), OutOfOrderTableProxy.validate] Access to a protected member _valida
te_out_of_order_table of a client class
tomlkit/container.py:1126: [W0212(protected-access), OutOfOrderTableProxy._merge_aot_fragment] Access to a protected mem
ber _map of a client class
tomlkit/container.py:1128: [W0212(protected-access), OutOfOrderTableProxy._merge_aot_fragment] Access to a protected mem
ber _map of a client class
tomlkit/container.py:1136: [W0212(protected-access), OutOfOrderTableProxy._merge_aot_fragment] Access to a protected mem
ber _body of a client class
tomlkit/container.py:1141: [W0212(protected-access), OutOfOrderTableProxy._merge_aot_fragment] Access to a protected mem
ber _body of a client class
tomlkit/container.py:1141: [W0212(protected-access), OutOfOrderTableProxy._merge_aot_fragment] Access to a protected mem
ber _body of a client class
tomlkit/container.py:1174: [C0415(import-outside-toplevel), OutOfOrderTableProxy.__setitem__] Import outside toplevel (i
tems.item)
tomlkit/container.py:1173: [R0912(too-many-branches), OutOfOrderTableProxy.__setitem__] Too many branches (14/12)
tomlkit/container.py:1227: [W0212(protected-access), OutOfOrderTableProxy._remove_table] Access to a protected member _b
```

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ody of a client class
tomlkit/container.py:1229: [W0212(protected-access), OutOfOrderTableProxy._remove_table] Access to a protected member _r
emove_at of a client class
tomlkit/container.py:1267: [W0212(protected-access), ends_with_whitespace] Access to a protected member _previous_item o
f a client class
************* Module tomlkit.parser
tomlkit/parser.py:1: [C0302(too-many-lines), ] Too many lines in module (1297/1000)
tomlkit/parser.py:103: [W0621(redefined-outer-name), Parser.__init__] Redefining name 'string' from outer scope (line 5)
tomlkit/parser.py:146: [W0622(redefined-builtin), Parser.consume] Redefining built-in 'min'
tomlkit/parser.py:146: [W0622(redefined-builtin), Parser.consume] Redefining built-in 'max'
tomlkit/parser.py:164: [W1113(keyword-arg-before-vararg), Parser.parse_error] Keyword argument before variable positiona
l arguments list in the definition of parse_error function
tomlkit/parser.py:259: [R1705(no-else-return), Parser._parse_item] Unnecessary "elif" after "return", replace only that
"elif" with "if"
tomlkit/parser.py:308: [R1723(no-else-break), Parser._parse_comment_trail] Unnecessary "elif" after "break", replace onl
y that "elif" with "if"
tomlkit/parser.py:292: [R0912(too-many-branches), Parser._parse_comment_trail] Too many branches (17/12)
tomlkit/parser.py:377: [R1720(no-else-raise), Parser._parse_key_value] Unnecessary "else" after "raise", remove the "els
e" and de-indent the code inside it
tomlkit/parser.py:433: [R1705(no-else-return), Parser._parse_simple_key] Unnecessary "else" after "return", remove the "
else" and de-indent the code inside it
tomlkit/parser.py:453: [W0212(protected-access), Parser._parse_quoted_key] Access to a protected member _t of a client c
lass
tomlkit/parser.py:454: [W0212(protected-access), Parser._parse_quoted_key] Access to a protected member _t of a client c
lass
tomlkit/parser.py:488: [R1705(no-else-return), Parser._parse_value] Unnecessary "elif" after "return", replace only that
 "elif" with "if"
tomlkit/parser.py:480: [R0911(too-many-return-statements), Parser._parse_value] Too many return statements (12/6)
tomlkit/parser.py:480: [R0912(too-many-branches), Parser._parse_value] Too many branches (19/12)
tomlkit/parser.py:480: [R0915(too-many-statements), Parser._parse_value] Too many statements (53/50)
tomlkit/parser.py:677: [W0212(protected-access), Parser._parse_array] Access to a protected member _s of a client class
tomlkit/parser.py:744: [R0911(too-many-return-statements), Parser._parse_number] Too many return statements (8/6)
tomlkit/parser.py:920: [R0916(too-many-boolean-expressions), Parser._parse_string] Too many boolean expressions in if st
atement (10/5)
tomlkit/parser.py:920: [R1720(no-else-raise), Parser._parse_string] Unnecessary "elif" after "raise", replace only that
"elif" with "if"
tomlkit/parser.py:946: [R1705(no-else-return), Parser._parse_string] Unnecessary "else" after "return", remove the "else
" and de-indent the code inside it
tomlkit/parser.py:999: [W0212(protected-access), Parser._parse_string] Access to a protected member _idx of a client cla
ss
tomlkit/parser.py:1004: [W0212(protected-access), Parser._parse_string] Access to a protected member _idx of a client cl
ass
tomlkit/parser.py:867: [R0912(too-many-branches), Parser._parse_string] Too many branches (25/12)
tomlkit/parser.py:867: [R0915(too-many-statements), Parser._parse_string] Too many statements (64/50)
tomlkit/parser.py:1006: [R0914(too-many-locals), Parser._parse_table] Too many local variables (24/15)
tomlkit/parser.py:1033: [R1720(no-else-raise), Parser._parse_table] Unnecessary "elif" after "raise", replace only that
"elif" with "if"
tomlkit/parser.py:1061: [R1720(no-else-raise), Parser._parse_table] Unnecessary "elif" after "raise", replace only that
"elif" with "if"
tomlkit/parser.py:1131: [R1723(no-else-break), Parser._parse_table] Unnecessary "else" after "break", remove the "else"
and de-indent the code inside it
tomlkit/parser.py:1156: [W0212(protected-access), Parser._parse_table] Access to a protected member _validate_out_of_ord
er_table of a client class
tomlkit/parser.py:1124: [R1702(too-many-nested-blocks), Parser._parse_table] Too many nested blocks (6/5)
tomlkit/parser.py:1006: [R0912(too-many-branches), Parser._parse_table] Too many branches (30/12)
tomlkit/parser.py:1006: [R0915(too-many-statements), Parser._parse_table] Too many statements (81/50)
tomlkit/parser.py:1007: [W0613(unused-argument), Parser._parse_table] Unused argument 'parent'
************* Module tomlkit._types
tomlkit/_types.py:41: [R0903(too-few-public-methods), WrapperType] Too few public methods (0/2)
tomlkit/_types.py:50: [W0223(abstract-method), _CustomList] Method '__delitem__' is abstract in class 'MutableSequence'
but is not overridden in child class '_CustomList'
tomlkit/_types.py:50: [W0223(abstract-method), _CustomList] Method '__getitem__' is abstract in class 'Sequence' but is
not overridden in child class '_CustomList'
tomlkit/_types.py:50: [W0223(abstract-method), _CustomList] Method '__len__' is abstract in class 'Sized' but is not ove
rridden in child class '_CustomList'
tomlkit/_types.py:50: [W0223(abstract-method), _CustomList] Method '__setitem__' is abstract in class 'MutableSequence'
but is not overridden in child class '_CustomList'
tomlkit/_types.py:50: [W0223(abstract-method), _CustomList] Method 'insert' is abstract in class 'MutableSequence' but i
s not overridden in child class '_CustomList'
tomlkit/_types.py:62: [W0223(abstract-method), _CustomDict] Method '__delitem__' is abstract in class 'MutableMapping' b
ut is not overridden in child class '_CustomDict'
tomlkit/_types.py:62: [W0223(abstract-method), _CustomDict] Method '__getitem__' is abstract in class 'Mapping' but is n
ot overridden in child class '_CustomDict'
tomlkit/_types.py:62: [W0223(abstract-method), _CustomDict] Method '__iter__' is abstract in class 'Iterable' but is not
 overridden in child class '_CustomDict'
tomlkit/_types.py:62: [W0223(abstract-method), _CustomDict] Method '__len__' is abstract in class 'Sized' but is not ove
rridden in child class '_CustomDict'
tomlkit/_types.py:62: [W0223(abstract-method), _CustomDict] Method '__setitem__' is abstract in class 'MutableMapping' b
ut is not overridden in child class '_CustomDict'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__abs__' is abstract in class 'Complex' but is not ov
```

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erridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__add__' is abstract in class 'Complex' but is not ov
erridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__and__' is abstract in class 'Integral' but is not o
verridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__ceil__' is abstract in class 'Real' but is not over
ridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__eq__' is abstract in class 'Complex' but is not ove
rridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__floor__' is abstract in class 'Real' but is not ove
rridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__floordiv__' is abstract in class 'Real' but is not
overridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__int__' is abstract in class 'Integral' but is not o
verridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__invert__' is abstract in class 'Integral' but is no
t overridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__le__' is abstract in class 'Real' but is not overri
dden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__lshift__' is abstract in class 'Integral' but is no
t overridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__lt__' is abstract in class 'Real' but is not overri
dden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__mod__' is abstract in class 'Real' but is not overr
idden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__mul__' is abstract in class 'Complex' but is not ov
erridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__neg__' is abstract in class 'Complex' but is not ov
erridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__or__' is abstract in class 'Integral' but is not ov
erridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__pos__' is abstract in class 'Complex' but is not ov
erridden in child class '_CustomInt'
tomlkit/_types.py:74: [W0223(abstract-method), _CustomInt] Method '__pow__' is abstract in class 'Integral' but is not o
verridden in child class '_CustomInt'
```

Showing the first 100 of 234 lines. Full file: `week6\pylint.txt`.

CrossHair symbolic-execution counterexample

Source: week6\symbolic\crosshair_output.txt

```
ch_date.py:10: error: false when calling date_never_returns_datetime('0001-10-30 00:00') (which returns False)
```

CrossHair contract

Source: week6\symbolic\ch_date.py

```
"""CrossHair contracts: api.date() should never return a datetime."""
import datetime as _dt
import re
import tomlkit
from tomlkit._utils import parse_rfc3339

def date_never_returns_datetime(raw: str) -> bool:
    """
    pre: re.fullmatch(r"[0-9]{4}-[0-9]{2}-[0-9]{2}[T ][0-9]{2}:[0-9]{2}", raw)
    post: _
    """
    try:
        d = tomlkit.date(raw)
    except ValueError:
        return True
    return not isinstance(d, _dt.datetime)
```

Assertion audit summary

Source: week6\assertion_audit.csv

Value	Count	Share
STRONG	209	68.5%
WEAK	48	15.7%
PARTIAL	47	15.4%
NO-ASSERT (smoke)	1	0.3%
Total	**305**	**100.0%**

305 rows summarised by `verdict`.

AI-only coverage summary

Source: week6\cov_ai.json

Metric	Value
Statements	2,671 / 3,268
Statement coverage	81.73%
Branches	831 / 1,254
Branch coverage	66.27%
Combined coverage	77.44%
Excluded lines	118

Maintainers' existing coverage summary

Source: week6\cov_existing.json

Metric	Value
Statements	2,924 / 3,268
Statement coverage	89.47%
Branches	1,042 / 1,254
Branch coverage	83.09%
Combined coverage	87.70%
Excluded lines	118

AI test IDs used by the RTM

Source: week6\rtm\ai_test_ids.txt

```
388 tests collected in 0.07s
test_ai_api.py::TestContainerFactories::test_aot
test_ai_api.py::TestContainerFactories::test_array_default_empty
test_ai_api.py::TestContainerFactories::test_array_from_string
test_ai_api.py::TestContainerFactories::test_array_rejects_non_array
test_ai_api.py::TestContainerFactories::test_inline_table
test_ai_api.py::TestContainerFactories::test_super_table_renders_dotted_header
test_ai_api.py::TestContainerFactories::test_table
test_ai_api.py::TestDateTimeFactories::test_date
test_ai_api.py::TestDateTimeFactories::test_date_rejects_datetime
test_ai_api.py::TestDateTimeFactories::test_datetime_rejects_time
test_ai_api.py::TestDateTimeFactories::test_datetime_with_offset
test_ai_api.py::TestDateTimeFactories::test_invalid_string
test_ai_api.py::TestDateTimeFactories::test_time
test_ai_api.py::TestDateTimeFactories::test_time_rejects_date
test_ai_api.py::TestDocument::test_document_build
test_ai_api.py::TestDocument::test_document_empty
test_ai_api.py::TestDumps::test_dumps_nested_dict_creates_table
test_ai_api.py::TestDumps::test_dumps_plain_dict
test_ai_api.py::TestDumps::test_dumps_rejects_non_mapping
test_ai_api.py::TestDumps::test_dumps_sort_keys
test_ai_api.py::TestDumps::test_dumps_sort_keys_on_document
test_ai_api.py::TestDumps::test_dumps_table_item
test_ai_api.py::TestEncoders::test_encoder_must_return_item
```

Showing the first 25 of 307 lines. Full file: `week6\rtm\ai_test_ids.txt`.

The AI-only HTML coverage report is in `week6/htmlcov_ai/`; the raw JSON is `week6/cov_ai.json`. These are evidence, not printed page-for-page.

Raw evidence inventory

The pack prints the readable report and key summaries. Large raw artefacts are kept in the repository and listed here so a marker can verify them without paging through thousands of lines. Items tagged **[AI-only]** are deliberately not printed: they are the AI baseline's raw output, not the final hybrid result.

File	Size	Tag
week6\assertion_audit.csv	25 KB	
week6\cov_ai.json	388 KB	[AI-only]
week6\cov_existing.json	388 KB	
week6\pylint.txt	29 KB	
week6\radon_cc.json	146 KB	
week6\radon_cc.txt	1 KB	

File	Size	Tag
week6\radon_mi.txt	387 B	
week6\README.md	1 KB	
week6\rtm\ai_test_ids.txt	17 KB	
week6\rtm\build_rtm.py	7 KB	
week6\rtm\rtm_week6.csv	6 KB	
week6\scripts\assert_audit.py	3 KB	
week6\scripts\gap_by_function.py	1 KB	
week6\symbolic\ch_date.py	459 B	
week6\symbolic\crosshair_output.txt	112 B	